

ZHENZHUO LI

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EDUCATION

Hong Kong University of Science and Technology (Guangzhou) BSc, Data Science and Big Data Technology 2023 – 2027 (Expected)

- **CGA:** 3.807/4.3 | **Major CGA:** 3.837 | Dean's List (3 semesters: 2023-24 Spring, 2024-25 Fall, 2024-25 Spring)
- **Core Coursework:** Machine Learning (A-), Deep Learning (B+), Introduction to AI (A+), Design and Analysis of Algorithms (A), Advanced Algorithms and Data Structures (A+), Database Management Systems (A+), Cloud Computing and Big Data Systems (A-), Applied Statistics (A-), Linear Algebra (A+), Discrete Mathematics (A), Honors Calculus I/II (A+/A+)
- **Exchange:** Osaka University, Japan (2025 Fall)

PUBLICATIONS

- Y. Xu, J. Hao, A. Liu, **Z. Li**, S. Meng, S. Yuan, G. Zhang, "LENS: Large Pre-trained Transformer for Exploring Financial Time Series Regularities," in *Proceedings of the 6th ACM International Conference on AI in Finance (ICAIF '25)*, pp. 771–778, 2025. DOI: 10.1145/3768292.3770349. (Earlier version: *PLUTUS*, arXiv:2408.10111.)

PROFESSIONAL EXPERIENCE

Xuanyuan Private Fund Management (Guangdong) Co., Ltd. Quantitative Investment Modeling Research Intern 2026.07 – Present

Guangzhou, China

- **Independently responsible** for research on neural feature and factor discovery from **Level-2 equity-market data**, covering data construction, model development, signal evaluation, and trading-decision analysis
- Built scalable pipelines for multi-year minute-level market panels: temporal and cross-asset coordinate alignment, memory-mapped storage, parallel I/O, sample filtering, and model-ready dataset construction
- Developed **leakage-aware evaluation** procedures for intraday and cross-day signals: information coefficient, temporal stability (RPS), entry timing (Buy-Now-vs-Wait), and sample-selection audits
- Investigating neural and tree-based approaches (LightGBM, GRU, XGBoost) for market exposure and intraday entry decisions; research code and findings are currently under internal review

Shanghai Yunyan Private Fund Management Co., Ltd. AI Investment Research Intern 2026.05 – 2026.07

- Conducted AI-assisted investment research on **semiconductor and AI infrastructure sectors**; built workflows to collect and synthesize heterogeneous information from company disclosures, industry research, expert commentary, and open-source channels
- Evaluated investment hypotheses through multi-source evidence comparison, sector and company analysis, and market feedback
- Explored LLM-agent and quantitative approaches for information screening and signal validation, which motivated my interest in the **reliability and verification of AI agents** operating under noisy, heterogeneous, dynamically changing information

RESEARCH EXPERIENCE

LENS: Large Pre-trained Transformer for Financial Time Series Co-author (4th) 2024.06 – 2024.10

Research Assistant | Supervisor: Prof. Guang Zhang, HKUST(GZ) FinTech

- Developed parallel and memory-efficient preprocessing pipelines for terabyte-scale financial trading data; achieved **up to 13× faster data cleaning** and **up to 6× faster statistical aggregation** in internal benchmarks through parallelization and chunk-based processing
- Co-authored the resulting paper **published at ACM ICAIF 2025** (pp. 771–778); earlier version available as *PLUTUS* (arXiv:2408.10111)

Undergraduate Researcher | Supervisor: Prof. Zeke Xie, HKUST(GZ) DSA

- Led an independent study of failure modes in multi-region diffusion-based generation; developed a **training-free framework** for spatially controlled synthesis
- Conducted theoretical analysis and controlled experiments, and completed a full first-author manuscript through **one peer-review cycle**
- The project concluded after submission; it demonstrated my ability to lead open-ended research from problem formulation to method design, implementation, and manuscript preparation

SELECTED PROJECTS

AI-Assisted Quantitative Research SystemIndependent Research and Development2026.01 – 2026.06

- **Prototyped** an AI-assisted research workflow connecting timestamped information collection, factor construction, signal screening, and walk-forward evaluation
- Implemented a reusable factor-analysis module covering **39 price-volume and technical signals** with IC analysis, decay diagnostics, turnover analysis, and transaction-cost assumptions
- **Experimented with** LightGBM and XGBoost signal models, ensemble combination, and rule-based position sizing across historical market regimes (2023–2025 walk-forward experiments)
- The project provided an initial research framework later refined through institutional Level-2 data and leakage-aware signal research at Xuanyuan Investment

AWARDS & HONORS

- **ASC World University Supercomputer Competition 2026 — First Prize** (Team Member). Contributed I/O optimization for large-model inference and quantum-circuit simulation optimization.
- **HKUST(GZ) TradeMaster Cup Quantitative Trading Competition — Team 5th Place (Team Leader)**. Quantitative factor backtesting and trading strategy competition.
- **Lizhi Innovation Talent Scholarship (INFH)**, HKUST(GZ) — jointly awarded by the Information Hub and Lizhi Group for innovative research potential (2025)
- **MCM/ICM Mathematical Contest in Modeling — Honorable Mention** (2025)
- **ASC World University Supercomputer Competition 2025 — Second Prize** (Team Member)
- **Dean’s List**, HKUST(GZ) — 3 semesters (2023-24 Spring, 2024-25 Fall, 2024-25 Spring)
- **Admission Scholarship (First Class)**, HKUST(GZ) — Merit-based scholarship (2023)
- **Chinese Physics Olympiad (CPHO)** — Provincial First Prize, 38th and 39th editions (2021, 2022)

LEADERSHIP & ACTIVITIES

- **Student Vice Secretary**, HKUST(GZ) Youth League Committee (2024.05 – 2025.05)
- **Committee Member**, HKUST(GZ) Student Affairs Committee (2024.07 – 2025.06)
- **Member**, Nansha Harbor Toastmasters (Dynamic Leadership path; hosting and structured-presentation roles)
- **Delegate**, Boao Forum for Asia Youth Summit (2024.11)
- **Participant**, ABBA Youth Association / Prudential Elite Internship Camp (2025.02)

TECHNICAL SKILLS

Programming & Data: Python, C++, SQL; NumPy, Pandas, SciPy, scikit-learn

Machine Learning: PyTorch, LightGBM, XGBoost, Transformers, scikit-learn

Quantitative Research: Factor and representation discovery, information-coefficient and decay analysis, leakage-aware evaluation, walk-forward validation, transaction-cost modeling, signal-to-decision modeling

Data & Systems: Linux, Git, large-scale array processing (mmap), CUDA, MPI, Slurm

Languages: Chinese (native), English (full professional proficiency; undergraduate program taught in English), Japanese (JLPT N1)

Research Interests: Financial time-series representation learning; foundation models for financial data; reliable machine learning for financial markets — signal reliability under non-stationarity, leakage-aware evaluation, and robust transfer; neural feature discovery from market microstructure (Level-2) data